

Chapter 3. **GROUPS**

3.4 Homomorphisms and isomorphisms



Today we will

- introduce group homomorphisms more formally
- see some examples

What is a group homomorphism?

Here, we will study maps from one group onto another group that preserve the group operation. Roughly speaking, an isomorphism between two groups is a way to transform one group into the other one via a one-to-one map. In such a case, we say that such two groups are isomorphic. More precisely: let G_1 and G_2 be two groups. An **isomorphism** from G_1 to G_2 is a bijective correspondence φ from G_1 to G_2 satisfying the following property: for every pair of elements a and b in G_1 ,

$$\text{if } \varphi(a) = a' \text{ and } \varphi(b) = b', \text{ then } \varphi(ab) = a'b' = \varphi(a)\varphi(b) \quad (*)$$

What does (*) say?

To compute $\varphi(ab)$, we can first find ab and then apply φ (to ab), or we can compute $\varphi(a)$ and $\varphi(b)$, and multiply these two elements.

In §3.1.1 we called two groups isomorphic if they have the same Cayley table except for notation. Observe that this terminology is consistent with the one that we have just introduced. If $\varphi : G_1 \rightarrow G_2$ is an isomorphism, then using $(*)$ we can see that φ transform the Cayley table of G_1 into the Cayley table of G_2 ; in fact:

G_1	b		G_2	b'
	$\vdots$			$\vdots$
a	$\dots$ <u>ab</u>	For every x replace x by $\varphi(x)$	a'	$\dots$ <u>$a'b'$</u>

$ab \rightarrow \varphi(ab) \stackrel{(*)}{=} \varphi(a)\varphi(b) = a'b'$

The idea behind two isomorphic groups is the following: If G_1 and G_2 are isomorphic, then we can say that they are indeed the same, except that their elements have different labels. The group G_1 becomes G_2 after relabelling its elements; the map which does the relabelling is precisely an isomorphism from G_1 to G_2 .

Example

$\mathbb{Z}^*$	1	-1
1	1	-1
-1	-1	1

φ

$\mathbb{Z}_2$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{0}$

$\bar{1} + \bar{1}$

$$\varphi(1) = \bar{0}$$

$$\varphi(-1) = \bar{1}$$

$$\varphi((-1)(-1)) = \varphi(-1)\varphi(-1) = \bar{1} + \bar{1} = \bar{0}$$

Definitions 3.36. Let $(G_1, \cdot_1)$ and $(G_2, \cdot_2)$ be two groups. A mapping $\varphi : (G_1, \cdot_1) \rightarrow (G_2, \cdot_2)$ is called a **group homomorphism** if

$$\varphi(a \cdot_1 b) = \varphi(a) \cdot_2 \varphi(b), \quad \leftarrow$$

for all $a, b \in G_1$. A **monomorphism** is a one-to-one homomorphism, while a surjective homomorphism is called an **epimorphism**. Homomorphisms which are bijectives (one-to-one and onto) are said to be **isomorphisms**. When an isomorphism exists from G_1 to G_2 we say that G_1 **is isomorphic to** G_2 , and we write $G_1 \cong G_2$. Isomorphisms from a group to itself are called **automorphisms**.

How can we find out whether two groups are isomorphic?

1. Use your intuition to come up with a map $\varphi : G_1 \rightarrow G_2$, which looks like it could be an isomorphism. The more exercises you do, the easier this first step will become.
2. Check that φ is bijective.
3. Take two arbitrary elements a and b in G_1 , compute $\varphi(a)$, $\varphi(b)$ and $\varphi(ab)$ and check that the equality $\varphi(ab) = \varphi(a)\varphi(b)$ holds.

Examples 3.37.

① The additive group $\mathbb{R}$ is isomorphic to the multiplicative group $\mathbb{R}^+$.

$$\varphi: (\mathbb{R}, +) \longrightarrow (\mathbb{R}^+, \cdot)$$

•) φ bijective

•) $\varphi(a+b) = \varphi(a)\varphi(b)$ for all $a, b \in \mathbb{R}$.

Let $\varphi(x) = e^x$ for all $x \in \mathbb{R}$. It is bijective because it has inverse ($\ln$ the inverse).

For a and $b \in \mathbb{R}$, we have

$$\varphi(a+b) = e^{a+b} = e^a e^b = \underline{\varphi(a)\varphi(b)}$$

φ is a group isomorphism.

$$\textcircled{1} \quad \varphi: (\mathbb{Z}, +) \longrightarrow (\mathbb{Z}, +)$$

$$a \longmapsto \varphi(a) = 3a$$

•) φ is injective:

$$\varphi(a) = \varphi(b) \Rightarrow 3a = 3b \Rightarrow 3(a-b) = 0$$

$$\Rightarrow a - b = 0 \Rightarrow a = b$$

•) φ is not surjective:

There is no integer a such that $\varphi(a) = 5$ because the equation $3a = 5$ does not have a solution in the integers.

•) φ is a group homomorphism:

$$\underline{\varphi(a+b)} = 3(a+b) = 3a + 3b = \underline{\varphi(a) + \varphi(b)}$$

We have proved that φ is a group monomorphism.

② The determinant map

$$\begin{aligned}\varphi: GL_n(\mathbb{R}) &\longrightarrow \mathbb{R}^* \\ A &\longmapsto \det(A)\end{aligned}$$

.) φ is a group homomorphism:

$$\begin{aligned}\varphi(AB) &= \det(AB) = \det(A)\det(B) \\ &= \varphi(A)\varphi(B), \text{ for all } A, B \in GL_n(\mathbb{R}).\end{aligned}$$

.) Is φ a monomorphism? No

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\det(A) = -1 = \det(B)$$

$$\text{but } A \neq B.$$

•) Is φ an epimorphism?

For every $0 \neq a \in \mathbb{R}$, there exists $\textcircled{\text{yes}}$
a matrix A such that $\varphi(A) = \det(A) = a$

$$A = \begin{pmatrix} a & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{pmatrix}$$

$$\det(A) = a.$$

We have proved that φ is group
epimorphism.
